SANJOY DAS

Data Analyst | Python | SQL | Power BI | Machine Learning | Time-Series Forecasting

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PROFESSIONAL SUMMARY

Entry-level Data Analyst with a B.E. in Electrical Engineering, two data analyst internships and four end-to-end analytics projects covering forecasting, predictive maintenance, anomaly detection and customer churn. Skilled in Python (Pandas, Scikit-learn, Statsmodels), Streamlit, Power BI, SQL and Advanced Excel, with a record of turning raw data into dashboards, predictive models and executive-ready recommendations. Brings industrial DCS and process-plant experience (CFBC boiler) to sensor, operations and performance analytics. Seeking a Data Analyst role to support data-driven business decisions.

TECHNICAL SKILLS

Programming & Libraries: Python (Pandas, NumPy, Scikit-learn, Statsmodels, TensorFlow/Keras), SQL (SELECT, JOIN, GROUP BY, UNION)

Data Visualization & BI: Power BI (DAX, Power Query, Data Modeling), Streamlit, Plotly, Matplotlib, Seaborn, Advanced Excel (Pivot Tables, Dashboards, Power Query)

Machine Learning: Random Forest, Gradient Boosting, Logistic Regression, K-Means Clustering, PCA, Isolation Forest, Autoencoder, Count Vectorizer; Model Evaluation (RMSE, MAE, MAPE, Accuracy, ROC-AUC, Average Precision)

Time-Series Forecasting: ARIMA, SARIMA, Holt-Winters Exponential Smoothing, Seasonal Decomposition, Walk-Forward Backtesting, Rolling-Origin Cross-Validation, Prediction Intervals

Statistics: Hypothesis Testing (Chi-Square, Z-Test, T-Test, ANOVA), Correlation Analysis, Confidence Intervals

Analytics Practices: Data Cleaning & Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, KPI Analysis, Anomaly Detection, Churn & Customer Segmentation, Dashboard Design, Executive Reporting

EXPERIENCE

Data Analyst Intern | Unified Mentor Pvt Ltd — 15 Mar 2026 – Sep 2026

- Built four end-to-end analytics projects (customer retention, care-load forecasting, IoT predictive maintenance, churn segmentation), each delivered as an interactive multi-tab Streamlit dashboard, deployed live and published on GitHub, with dynamically computed insights.
- Cleaned, engineered features and performed EDA in Python (Pandas, NumPy) on datasets ranging from 10K to 100K records.
- Developed and validated predictive models (Gradient Boosting churn classifier, Random Forest forecasting, Isolation Forest anomaly detection) using time-based splits, cross-validation and backtesting.
- Produced a research paper and executive summaries that translate technical results into recommendations for non-technical stakeholders.

Data Analyst Intern | InlignX Global Pvt. Ltd. — Nov 2025 – Feb 2026

- Cleaned and preprocessed data in Python (Pandas, NumPy), resolving missing values, duplicates and inconsistencies.
- Developed interactive Power BI dashboards to track KPIs and performance metrics.
- Wrote and optimized SQL queries to extract, filter and analyze structured data from databases.
- Conducted statistical analysis with Statsmodels to identify trends, correlations and key insights.

DCS Engineer | Orissa Steel and Power Private Ltd. — Jun 2025 – Present

- Monitored process parameters of a 180 TPH CFBC boiler on the DCS, analyzing operational and performance data to track efficiency trends and flag anomalies.
- Prepared structured reports and summary dashboards to support data-driven decision-making.

PROJECTS

HHS UAC Program: Predictive Forecasting of Care Load and Placement Demand | *Python, Streamlit, Scikit-learn, Statsmodels, Plotly*

GitHub: github.com/sanjoydas087/hhsdashboard | **Live App:** hhsdashboard.streamlit.app/

- Built a 5-module Streamlit dashboard on 1,061 days of program data to forecast care load, intake-versus-discharge imbalance and short-term discharge demand, with an early-warning capacity alert system and what-if scenario simulation.
- Benchmarked 7 models (Naive, Moving Average, Holt-Winters, ARIMA, SARIMA, Random Forest, Gradient Boosting) on engineered lag, rolling-window and calendar features; selected Random Forest (RMSE 6.6, 99.8% accuracy on a 60-day chronological holdout).
- Quantified forecast uncertainty with prediction intervals, 5-fold rolling-origin cross-validation and a 40-origin walk-forward backtest, which showed long-horizon error growing faster than the standard square-root-of-time assumption.

- Uncovered a structural break: average daily care load fell about 71% (2023 to 2025) with intake down over 90% in two quarters. Delivered a research paper and executive summary with capacity-planning recommendations.

Predictive Maintenance and Anomaly Detection: Manufacturing IoT (Thales Group case study) | *Python, Scikit-learn, TensorFlow/Keras, Streamlit*

GitHub: github.com/sanjoydas087/PredictiveMaintenanceThalesSecondProject | **Live App:** predictivemaintenancethalessecondproject-hikyetx4b4q5cwmbpxsrt.streamlit.app/

- Analyzed 100K+ IoT sensor records across 50 machines to detect abnormal equipment behavior and support predictive maintenance planning.
- Built anomaly detection models with Isolation Forest and an Autoencoder, and created a business-readable Anomaly Index to rank machines by severity.
- Fixed a many-to-many join duplication when combining machine data files by aligning records by position after identical sorting, protecting data integrity.
- Delivered a multi-tab Streamlit dashboard with an industrial theme and dynamically computed findings for maintenance monitoring.

Customer Segmentation and Churn Pattern Analytics: European Banking | *Python, Scikit-learn, Streamlit, Plotly*

GitHub: github.com/sanjoydas087/Banchurnappfourth | **Live App:** banchurnapp.streamlit.app/

- Built a multi-tab Streamlit dashboard segmenting bank customers and analyzing churn patterns to identify high-risk groups and key churn drivers.
- Trained a Gradient Boosting churn classifier (ROC-AUC about 0.86), adding a predictive scoring layer on top of descriptive analytics.
- Delivered key findings and an executive summary with targeted retention recommendations.

ECB Customer Retention Analytics Dashboard: European Banking | *Python, Streamlit, Plotly, Pandas*

GitHub: github.com/sanjoydas087/Customer-Retention-Dashboard | **Live App:** customer-retention-dashboard-ytytyubuub.streamlit.app/

- Analyzed 10,000 European bank customer records to uncover churn patterns and retention drivers, presented through an interactive Streamlit dashboard with KPI cards and dynamic insights.
- Structured the analysis around retention KPIs so business teams can spot at-risk customer groups and prioritize retention actions.

Executive Sales Performance and Operations Analytics Dashboard | *Power BI, DAX*

- Developed a 4-page executive dashboard analyzing $2.73M in sales revenue over 3+ years, with 15+ interactive visuals including maps, KPI cards, time-series and drill-down reports.
- Created DAX measures showing a 55% profit margin with Technology as the top category; segmentation showed the Consumer segment contributing 42% of revenue; built budget-versus-actual variance analysis across 7 cost centers.

Banking Customer Analytics Dashboard | *Power BI*

- Analyzed 4,000+ banking customers with total balances exceeding 160M; tracked monthly acquisition trends, demographics by age, gender, job and region, and identified high-value customers through balance distribution analysis.

High-Value Transaction Prediction | *Python, Scikit-learn*

- Built an end-to-end classification pipeline (ColumnTransformer, StandardScaler) with a Logistic Regression model evaluated on Accuracy, AUROC and Average Precision.

IMDB Data Analysis | *SQL Server Management Studio*

- Designed the database and tables and wrote SQL queries (UNION, aggregations) to rank top actors by appearances and top directors by average rating, and to count movies per year and genre.

Zomato Restaurant Analytics | *Python, Power BI*

- Cleaned and transformed a real-world Zomato dataset, ran EDA on ratings, pricing and popularity using Matplotlib and Seaborn, and built an interactive Power BI dashboard.

EDUCATION

B.E. in Electrical Engineering | Jadavpur University 2000

CERTIFICATIONS

- Advanced Certification in Data Analytics, IRA Skills
- Global Certificate in Environmental Data & GIS
- Certified Energy Manager and Energy Auditor (CEM, CEA)

LANGUAGES

English, Hindi, Bengali